

Reading Scene Text for Visual Question Answering

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1 Introduction

Many visual question answering systems can describe objects, scenes, and simple spatial relations, but they still fail when the answer depends on text inside the image. Text is common in real visual environments: product labels, street signs, receipts, medical device screens, drug packaging, forms, dashboards, and instructions often carry the information needed to answer a question. A model that ignores or misreads this text may produce an answer that is fluent but wrong.

This project studies that failure mode using TextVQA, a benchmark whose questions require reading and reasoning over scene text [5]. Each example pairs an image with a natural-language question and multiple human answers. The model must identify which visible text is relevant, read it correctly, and return a concise answer that matches the annotators’ responses. We use the predefined TextVQA train and validation splits from Hugging Face, with 34.6k training examples and 5k validation examples.

This capability is relevant to medicine and health-adjacent workflows because visual information is often mixed with text: medication labels, monitor readouts, intake forms, appointment slips, nutrition labels, and device instructions. This project does not train on protected clinical data, but TextVQA provides a controlled benchmark for a related capability: answering questions from images where written text carries the key evidence.

Our goal is to measure how well a modern open-source vision-language model handles this text-centric setting and then improve it with a reproducible adaptation method. We use Qwen2.5-VL-3B-Instruct as the base model because it is small enough for accessible multi-GPU fine-tuning while still being designed for document, chart, and text-rich visual understanding [1]. The main improvement method is LoRA fine-tuning, which freezes the pretrained model weights and trains low-rank adapter matrices in selected transformer modules [2]. The experiments compare zero-shot inference, LoRA fine-tuning, prompt-based OCR instructions, LoRA target-module choices, and image-resolution changes under the same validation split and scoring code.

2 Background

2.1 Text-Centric Visual Question Answering

TextVQA was introduced to evaluate VQA models on scene text reading, a capability beyond standard visual object recognition [5]. The original work showed that standard VQA systems struggled on questions grounded in text and proposed Look, Read, Reason, and Answer (LoRRA), a model that combines visual features, OCR tokens, and question context. The benchmark remains useful for modern VLMs because the hard cases extend beyond OCR. The model must also decide which text matters, normalize short answers, and avoid hallucinating plausible but unseen words.

Evaluation follows the VQA-style multi-answer setup. Each prediction is compared against the set of human answers, and TextVQA soft accuracy gives partial credit based on annotator agreement. This is more appropriate than a single exact-match label because valid answers can vary in spelling, punctuation, and specificity. In this report, TextVQA soft accuracy is the primary metric, while exact match, token F1, BLEU, METEOR, and ROUGE-L provide secondary views of answer quality.

2.2 Modern Vision-Language Models

Recent VLMs differ in how they connect images to language models. BLIP-2 uses a frozen image encoder, a frozen language model, and a trainable Q-Former bridge to reduce the cost of vision-language pretraining [3]. LLaVA connects a vision encoder to a language model and improves instruction-following behavior through visual instruction tuning [4]. These designs helped move VLMs from task-specific architectures toward general-purpose visual assistants, but text-heavy images still stress their visual resolution, OCR alignment, and answer normalization.

Qwen2.5-VL is a more recent open-source VLM family with explicit emphasis on visual recognition, document parsing, localization, and dynamic-resolution processing [1]. Those properties make Qwen2.5-VL-3B-Instruct a strong fit for TextVQA: the model is compact enough for our compute budget, supports instruction-style prompting, and is built for the kinds of text-rich visual inputs that dominate this benchmark.

2.3 Parameter-Efficient Fine-Tuning

Full fine-tuning updates all model weights and can be costly for large VLMs. LoRA instead freezes the pretrained weights and inserts low-rank trainable matrices into selected linear layers [2]. This reduces memory use and checkpoint size while allowing the model to adapt to a downstream task. In this project, the vision side remains frozen, and LoRA adapters are applied to language-model attention and MLP projection modules. This choice matches the goal of improving answer generation and text-conditioned reasoning without retraining the whole multimodal backbone.

3 Methodology

3.1 Data Pipeline

The data pipeline downloads TextVQA from the Hugging Face dataset `lmms-lab/textvqa` and preprocesses the train and validation splits into local JSONL files. Each raw row is normalized into a common schema containing the question id, image id, question text, answer list, majority target answer, OCR tokens, and cached image path used during inference. Images are converted to RGB JPEGs and resized with a longest side of 448 pixels for the main runs. For training, each example is converted into a Qwen chat-style multimodal sequence. The user message contains the image and the prompt: “Question: {question}. Answer with the shortest correct text from the image.” The assistant message contains the majority human answer from the TextVQA answer list. This keeps the supervised target concise and close to the benchmark answer format.

3.2 Base Model and Input Formatting

The main model is `Qwen/Qwen2.5-VL-3B-Instruct`. The model is loaded with bfloat16 weights, scaled dot-product attention, and Qwen’s processor with the configured dynamic pixel range. For

distributed inference, each process maps the model to its local GPU. For distributed training without 4-bit quantization, the model is loaded without an automatic device map so that **Accelerate** and the Hugging Face **Trainer** can manage the process-local model placement.

Both training and inference use Qwen’s chat template and **qwen-vl-utils** to prepare image inputs. At inference time, the prompt template asks the model to answer only the question, and generation uses greedy decoding with a maximum of 32 new tokens. This decoding choice reduces variation across runs and matches the short-answer structure of TextVQA.

The codebase also includes **Salesforce/blip2-opt-2.7b** as a secondary model-comparison path. BLIP-2 is loaded with bfloat16 weights through the Hugging Face BLIP-2 generation API and processor. Its inference path uses a BLIP-2-specific prompt template and a cleanup step that removes echoed prompts or answer prefixes from OPT decoding outputs before scoring. This lets the experiments compare Qwen, a newer native multimodal instruction model, with BLIP-2, a frozen-vision/Q-Former/OPT bridge architecture.

3.3 LoRA Fine-Tuning

The improvement method is parameter-efficient LoRA fine-tuning. The vision side of the model is frozen, and LoRA adapters are inserted into the language-model attention and MLP projection layers: **q_proj**, **k_proj**, **v_proj**, **o_proj**, **gate_proj**, **up_proj**, and **down_proj**. This design lets the model adapt its text-conditioned reasoning and answer generation while avoiding the cost of updating the full VLM.

For BLIP-2, the same training configuration is reused where possible. The full BLIP-2 LoRA configuration targets OPT attention and feed-forward modules together with Q-Former query/value projections, while the Q-Former-only ablation targets only the bridge projections. This isolates the comparison to architecture and adapter placement while holding the dataset, optimizer schedule, and validation protocol fixed.

The LoRA hyperparameters follow common practice for instruction-tuned VLMs. We use rank $r=16$, alpha $\alpha=32$, and dropout 0.05 applied to the seven projection modules listed above. Rank 16 gives enough capacity to adapt all seven modules without saturating memory at our effective batch size; the $\alpha/r=2$ ratio is the conventional scaling so that each adapter update contributes stably during training; and the 0.05 dropout matches the value used in the original LoRA paper [2]. Optimization uses AdamW at learning rate 1×10^{-4} with a cosine schedule and 3% linear warmup, a typical starting point for LoRA on bf16 transformers that matches PEFT defaults. Training runs for two epochs in bfloat16 with a per-device batch size of 2 and four gradient-accumulation steps, giving an effective batch size of 32 across four GPUs. Sequences are capped at 2,048 tokens, gradient checkpointing is enabled so the 3B-parameter base model fits in memory, and the random seed is 42. A 512-example holdout from the end of the training split is reserved for training-time evaluation, leaving the official 5,000-example validation split for final reporting. The focus of this report is the model-comparison axis, so we hold these adapter hyperparameters fixed across runs and ablate target-module placement instead.

3.4 Evaluation Procedure

All reported validation runs use the same processed TextVQA validation split of 5,000 examples. Each prediction record preserves the question, the human answer list, the image reference, and the model output, and the same scoring code then computes metrics for the zero-shot, fine-tuned, and ablation runs.

The primary metric is TextVQA soft accuracy. Before scoring, predictions and answers are lowercased, stripped of punctuation, and normalized by removing English articles. A prediction receives credit according to the number of matching human answers, capped at one: $\min(\text{matches}/3, 1)$. Exact match and token-level F1 are included as additional answer-level metrics. BLEU, METEOR, ROUGE-1, ROUGE-2, and ROUGE-L provide secondary semantic-overlap measurements, and the evaluation code also supports an LLM-as-a-judge pass over a capped sample for qualitative similarity checking.

3.5 Reproducibility

The pipeline is organized into stage-specific entrypoints: dataset download and preprocessing, zero-shot inference, LoRA training, fine-tuned inference, ablations, and result generation. Each stage is driven by a versioned configuration file, and all generated artifacts (metrics, plots, predictions, qualitative examples) are kept separate from raw data and checkpoints. This separation makes every reported number traceable to a specific model configuration, LoRA configuration, training configuration, and prediction set.

4 Experimental Design

4.1 Splits and Controls

The official TextVQA test labels are not included in the local artifacts, so all reported model comparisons use the predefined 5,000-example validation split. The training split is used only for LoRA adaptation. During training, 512 examples are held out from the end of the training split for training-time evaluation; this holdout is separate from the official validation split used for final reporting.

The experiments are designed to isolate one change at a time where possible. Within each architecture, the zero-shot and fine-tuned runs use the same validation examples, answer normalization, and greedy decoding setup. The Qwen LoRA target-module ablation changes only the adapter placement. The OCR-instruction and OCR-hint ablations change only the prompt and do not retrain the model. The high-resolution ablation changes the image preprocessing and model pixel budget while reusing the trained full-LoRA Qwen adapter. The BLIP-2 runs use the same split and scoring code, but they use BLIP-2’s processor, prompt format, and architecture-specific LoRA targets.

4.2 Runs

Table 1: Baseline and main fine-tuned runs.

Run	Purpose	Controlled comparison
Qwen zero-shot	Baseline for Qwen’s pretrained behavior on TextVQA.	No training; answer-only prompt; 448-pixel preprocessing; greedy decoding.
Qwen full LoRA	Main Qwen improvement method.	Same validation split and decoding as zero-shot; LoRA on attention and MLP projection modules.
BLIP-2 zero-shot	Baseline for a Q-Former/OPT architecture.	No training; BLIP-2 prompt template; same validation split and metrics.
BLIP-2 full LoRA	Main BLIP-2 improvement method.	Same training hyperparameters as Qwen; adapters target OPT and Q-Former modules.

Table 2: Ablation runs.

Run	Purpose	Controlled comparison
Qwen attention-only LoRA	Tests whether MLP adapter targets matter.	Same training and evaluation setup as full LoRA, but adapters are limited to <code>q/k/v/o_proj</code> .
Qwen OCR prompt	Tests prompt-only improvement without parameter updates.	Same pretrained model and validation split as zero-shot, but the prompt explicitly asks the model to read visible text first.
Qwen OCR hint	Tests explicit OCR-token context.	Same pretrained model and validation split as zero-shot, but the prompt includes detected OCR tokens.
Qwen high-resolution	Tests whether text legibility is a bottleneck.	Same full-LoRA adapter, but validation images are preprocessed at 896 pixels and Qwen’s max pixel budget is increased.
BLIP-2 Q-Former-only	Tests whether adapting only the bridge is sufficient.	Same BLIP-2 setup, but LoRA is limited to Q-Former query/value projections.
BLIP-2 OCR hint	Tests OCR-token prompting for BLIP-2.	Same BLIP-2 zero-shot model, but the prompt includes detected OCR tokens.

4.3 Evaluation Metrics

All metrics are computed over the same normalized strings: $\tilde{x}$ denotes the prediction or reference after lowercasing, punctuation removal, and English-article removal (**a**, **an**, **the**). For one example, p is the model prediction and $A = \{a_1, \dots, a_K\}$ is the human answer list (typically $K = 10$); aggregate scores are the mean over the 5,000 validation examples unless stated otherwise.

4.3.1 TextVQA Soft Accuracy

The primary metric. It matches TextVQA’s multi-answer protocol by giving partial credit based on annotator agreement, so a prediction that several humans also gave receives full credit while a less-frequent variant receives a fraction. This is more appropriate than a single exact-match label

because valid TextVQA answers vary in spelling, casing, and short-form choice.

$$\text{SoftAcc}(p, A) = \min\left(\frac{|\{a \in A : \tilde{a} = \tilde{p}\}|}{3}, 1\right)$$

4.3.2 Exact Match

A strict answer-level metric: 1 if the normalized prediction equals the normalized form of any reference, else 0. It is more conservative than soft accuracy and useful as a sanity check that the model is not relying on partial-credit boundaries.

$$\text{EM}(p, A) = \mathbf{1}[\exists a \in A : \tilde{a} = \tilde{p}]$$

4.3.3 Token F1

A bag-of-tokens overlap score that rewards partially correct answers, for example when the prediction contains the right brand but adds an extra word. We take the maximum F1 over the reference set, where overlap is the multiset intersection of token counts.

$$P = \frac{|\tilde{p} \cap \tilde{a}|}{|\tilde{p}|}, \quad R = \frac{|\tilde{p} \cap \tilde{a}|}{|\tilde{a}|}, \quad \text{F1}(p, A) = \max_{a \in A} \frac{2PR}{P+R}$$

4.3.4 BLEU

Corpus-level n-gram precision (**sacrebleu**) of all predictions against the first listed reference, with a brevity penalty for short outputs. It captures whether the predicted phrase matches the reference at the n-gram level across the whole validation set, complementing per-example metrics that would otherwise be dominated by single-token answers.

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^4 w_n \log p_n\right), \quad w_n = \frac{1}{4}$$

where p_n is the corpus-level modified n-gram precision and BP is the brevity penalty.

4.3.5 METEOR

A unigram-alignment metric (NLTK with WordNet) that uses stem and synonym matching, then penalizes fragmented alignments. It is more lenient than BLEU when the prediction uses an equivalent surface form, which helps for short TextVQA answers where “coke” and “coca-cola” should be treated as close.

$$F_\alpha = \frac{PR}{\alpha P + (1 - \alpha)R}, \quad \text{METEOR} = F_\alpha \cdot (1 - \gamma \text{frag}^\beta)$$

with the standard $\alpha=0.9$, $\beta=3$, $\gamma=0.5$ and frag equal to the chunk-to-match ratio.

4.3.6 ROUGE-1, ROUGE-2, ROUGE-L

F-measures of unigram, bigram, and longest-common-subsequence (LCS) overlap between the prediction and the first reference (**rouge_score**, with stemming). ROUGE-1 and ROUGE-2 mirror

BLEU on the recall side, and ROUGE-L additionally rewards correct word order without requiring contiguous matches, which matters for multi-token answers like proper nouns.

$$\text{ROUGE-N} = \frac{2 P_n R_n}{P_n + R_n}, \quad \text{ROUGE-L} = \frac{2 P_{\text{lcs}} R_{\text{lcs}}}{P_{\text{lcs}} + R_{\text{lcs}}}$$

where P_n, R_n are n-gram precision and recall and $P_{\text{lcs}}, R_{\text{lcs}}$ use the LCS length over $|\tilde{p}|$ and $|\tilde{a}|$.

4.3.7 LLM-as-a-Judge

A semantic-equivalence score from `gpt-4.1-mini` computed on all 5,000 validation predictions per run. The judge sees the question, the human answer list, and the prediction, and returns a score in $[0, 1]$ for each item. It is included as a sanity check on cases where string-overlap metrics underrate paraphrases or near-synonyms (for example “cola” versus “coca-cola”). The reported number is the mean of the per-item scores returned by the judge:

$$\text{Judge} = \frac{1}{N} \sum_{i=1}^N s_i, \quad s_i \in [0, 1]$$

4.3.8 Error Taxonomy and Qualitative Examples

In addition to the scalar metrics above, the analysis uses an error taxonomy generated from the fine-tuned predictions, with categories for missed visible OCR, wrong text selected, partial match, empty answer, verbose or hallucinated, and other. This is necessary because aggregate accuracy cannot distinguish those failure modes, and the categories are referenced in the Results and Analysis section to explain where the model still fails after adaptation.

5 Results and Analysis

5.1 Main Quantitative Results

Table 3 reports validation metrics for the baseline and main fine-tuned runs. Qwen2.5-VL is much stronger than BLIP-2 on this TextVQA setup: Qwen zero-shot reaches 0.6858 soft accuracy, while BLIP-2 zero-shot reaches 0.0603. Qwen LoRA fine-tuning improves soft accuracy to 0.7453, an absolute gain of 5.95 percentage points over the Qwen zero-shot baseline, and is the best run across every metric in this subset.

Table 3: Validation performance for baseline and main fine-tuned runs (5,000 TextVQA examples).

Run	Soft acc.	Exact	Token F1	BLEU	METEOR	ROUGE-1	ROUGE-2	ROUGE-L	Judge
Qwen zero-shot	0.6858	0.7262	0.7718	14.22	0.4955	0.6159	0.1850	0.6155	0.7474
Qwen full LoRA	0.7453	0.7826	0.8104	46.64	0.5076	0.6456	0.1891	0.6451	0.7784
BLIP-2 zero-shot	0.0603	0.0680	0.1237	2.64	0.1057	0.0945	0.0370	0.0942	0.1417
BLIP-2 full LoRA	0.0537	0.0578	0.1718	1.05	0.1770	0.1224	0.0278	0.1219	0.3958

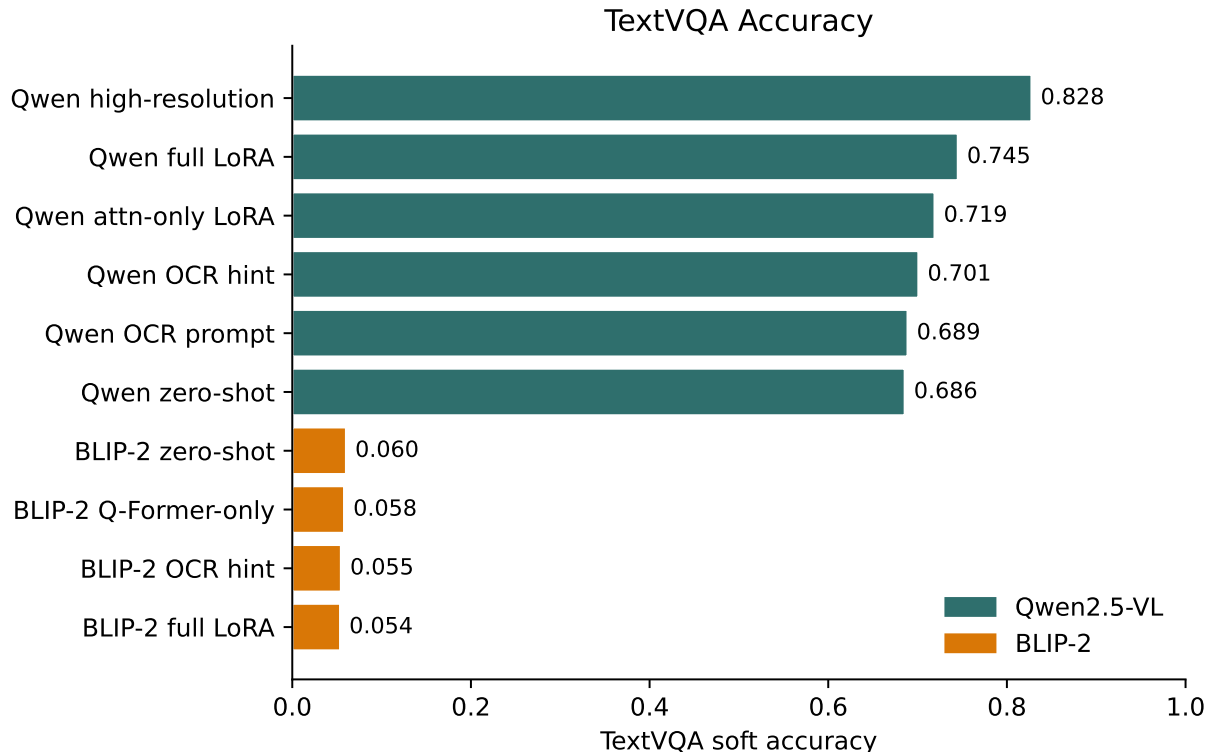


Figure 1: TextVQA soft accuracy across all 10 validation runs (baseline, main fine-tuned, and ablation), sorted by accuracy and color-coded by model family. Qwen2.5-VL runs are shown in teal, BLIP-2 runs in orange.

The main comparison shows that LoRA adaptation is useful for Qwen: full LoRA improves over Qwen zero-shot while keeping the same 448-pixel preprocessing, which indicates that task-specific answer formatting and text-conditioned reasoning can be learned through lightweight adapters without changing the visual front end. BLIP-2 performs far below Qwen in both zero-shot and LoRA settings, which suggests that this BLIP-2/OPT prompt-and-decoding setup is poorly matched to TextVQA’s short OCR-heavy answers.

The judge column makes the BLIP-2 failure mode visible. BLIP-2 LoRA reaches 0.3958 LLM-judge while only scoring 0.0537 soft accuracy, a gap roughly seven times larger than on any Qwen run. A direct look at the predictions confirms that most of this gap is decoding format rather than missing semantics: BLIP-2 LoRA outputs include echoed prompt fragments, repeated phrases (“*vermouth di marbella vermouth di marbella vermouth di marbella...*”), inserted unit strings (“*\$1.00 usd.00 usd.00 usd.00...*”), and over-specified product descriptions (“*mccall’s mark 70% abv bourbon whiskey*” for an image whose answer is “southern comfort”). The TextVQA normalizer expects a short answer string and is not robust to these patterns, so a prediction that contains the right token but is buried in surrounding text fails exact match, soft accuracy, and BLEU even when the underlying content is partially correct. OPT’s tendency to continue past the answer, combined with BLIP-2’s lack of an instruction-tuned chat template and our 32-token cap, means the LoRA adapter learns to produce on-topic but verbose continuations rather than short normalized answers. A judge that scores semantic equivalence ignores the formatting and recovers a much higher score, which is why we report it alongside string-overlap metrics. The implication is that closing the BLIP-2 gap on TextVQA likely needs answer-formatting supervision and stronger decoding constraints, not just more LoRA capacity.

5.2 Ablation Findings

Table 4 reports validation metrics for the ablation runs. The Qwen ablations separate four design choices: adapter target modules, prompt-only OCR instruction, OCR-token context, and image resolution. Attention-only LoRA improves over zero-shot by 3.34 percentage points but remains 2.61 points below full LoRA, which supports the choice to adapt both attention and MLP projection layers. The OCR prompt changes the model instruction without changing weights and gives only a small gain over zero-shot, from 0.6858 to 0.6890. Adding OCR-token hints improves more, to 0.7011, but still remains far below LoRA fine-tuning and high-resolution evaluation. The high-resolution run is the strongest validation setting overall, reaching 0.8277 soft accuracy, 0.8881 token F1, 49.62 BLEU, 0.5615 METEOR, 0.7028 ROUGE-1, 0.2096 ROUGE-2, 0.7021 ROUGE-L, and 0.8643 LLM-judge, an additional 8.25 points of soft accuracy over Qwen full LoRA.

Table 4: Validation performance for ablation runs (5,000 TextVQA examples).

Run	Soft acc.	Exact	Token F1	BLEU	METEOR	ROUGE-1	ROUGE-2	ROUGE-L	Judge
Qwen attention-only LoRA	0.7192	0.7556	0.7879	44.07	0.4966	0.6276	0.1871	0.6271	0.7541
Qwen OCR prompt	0.6890	0.7280	0.7716	13.42	0.4949	0.6179	0.1850	0.6175	0.7472
Qwen OCR hint	0.7011	0.7378	0.7788	14.87	0.4941	0.6213	0.1834	0.6210	0.7569
Qwen high-resolution	0.8277	0.8668	0.8881	49.62	0.5615	0.7028	0.2096	0.7021	0.8643
BLIP-2 Q-Former-only	0.0582	0.0654	0.1187	0.12	0.0789	0.0837	0.0056	0.0833	0.1087
BLIP-2 OCR hint	0.0547	0.0606	0.1116	1.74	0.1007	0.0885	0.0316	0.0882	0.1516

The high-resolution ablation has the largest effect. It uses the trained full-LoRA adapter but preprocesses validation images at 896 pixels and raises Qwen’s maximum pixel budget. Its accuracy gain is consistent with the task definition: many TextVQA answers are small numbers, brand names, or short printed phrases, so preserving more image detail directly improves the model’s chance of reading the relevant text. We note that this run conflates two changes, the longest-side preprocessing resolution (448→896) and Qwen’s per-image max-pixel cap, so the reported gain is the joint effect of more legible inputs and a larger visual token budget rather than either factor in isolation. The BLIP-2 ablations do not show the same pattern: full LoRA, Q-Former-only LoRA, and OCR-hint prompting all remain near 5-6% soft accuracy. This points to a model-architecture or formatting mismatch at a level that hyperparameter tuning is unlikely to close.

5.3 Question-Type Performance

Figure 2 breaks down fine-tuned performance by the first word of the question. The model performs best on several closed-form categories such as “does,” “is,” and “this,” and performs worse on some open-ended categories such as “when” and “what.” This pattern is expected: questions asking for dates, prices, names, or arbitrary text strings require precise OCR and region selection, while yes/no or object-identification questions often have a smaller answer space.

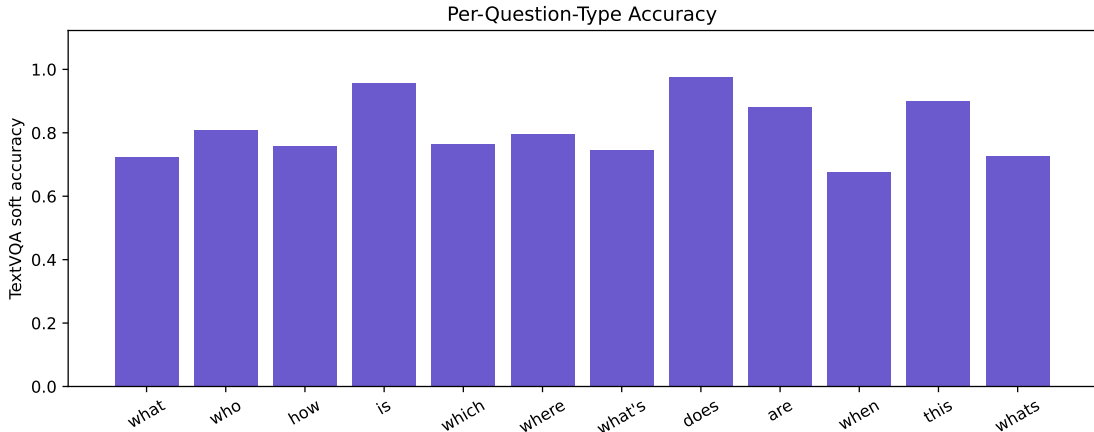


Figure 2: Fine-tuned TextVQA soft accuracy by question type.

5.4 Error Analysis

The error taxonomy in Table 5 was generated from the full-LoRA validation predictions. The largest non-correct group is “other,” which contains heterogeneous failures that are not captured by the narrower rules. Among named failure modes, missed visible OCR is the dominant issue, accounting for 377 examples. Wrong text selection and partial matches are smaller but important because they show that the model often sees some relevant text but either chooses the wrong region or returns an incomplete answer.

Table 5: Error taxonomy for the full-LoRA validation predictions.

Category	Count	Share
Correct	3913	78.26%
Other	537	10.74%
Missed visible OCR	377	7.54%
Partial match	113	2.26%
Wrong text selected	51	1.02%
Empty answer	6	0.12%
Verbose or hallucinated	3	0.06%

Qualitative examples support the same interpretation. The model correctly answers short text queries such as camera brands, city names, beer types, and ages when the relevant text is legible and well localized. Failures include reading a displayed time as “10:08” when the annotated answer is “5:41,” predicting a plausible watch brand when the answer set includes “AP” or “unanswerable,” and selecting the wrong name or product label in cluttered scenes. These examples show that the remaining errors go beyond answer formatting. They often require better OCR resolution, stronger grounding to the queried region, or explicit uncertainty when the text cannot be read.

Regrouping the taxonomy along an OCR-versus-reasoning axis sharpens this picture. Missed visible OCR (377), wrong text selected (51), and partial match (113) sum to 541 of the 1,087 errors (49.8%); these are perception-side failures where the relevant text exists but is misread, mis-localized, or truncated. Empty (6) and verbose-or-hallucinated (3) outputs are decoding-side issues. The remaining 537 “other” cases are heterogeneous, but spot checking shows they often involve correct OCR followed by question-conditioned reasoning that misfires, such as choosing among multiple visible items, comparing numbers on a label, or resolving spatial references like

“second from the left.” The 8.25-point lift from high-resolution evaluation, which only changes perception, is consistent with this split: roughly half of the LoRA model’s errors are OCR-bound and respond directly to more legible inputs, while a meaningful tail still needs stronger question-grounded reasoning that resolution alone cannot fix.

5.5 Qualitative Findings

Six representative validation examples are reproduced as image-and-prediction panels in Appendix B. Here we summarize the qualitative patterns they show.

On short brand-name questions where the relevant text is clearly displayed (Q1: bottle brand, Q6: bank sponsor), Qwen reads the answer correctly across most variants, while BLIP-2 substitutes generic objects or hallucinated brand strings.

On fine-grained perceptual cases (Q2: clock time, Q3: small bottle-label word, Q4: small price tag), every Qwen variant misreads the visible text. The high-resolution ablation does not always recover the right answer either: on Q2 it still predicts the wrong time, and on Q6 it actually substitutes “wells fargo” for the correct “charles schwab.” This suggests these failures need stronger question-conditioned grounding rather than only more pixels.

The OCR-hint prompt occasionally pulls Qwen toward a nearby but wrong token (“Dewar’s” for the second bottle from the left, “Aperitivo” instead of “barbieri”), showing that prefixing detected OCR tokens can mislead the model when several plausible tokens are visible.

BLIP-2 LoRA outputs are typically on-topic but verbose, with repeated phrases and inserted units (Q1, Q3, Q4, Q6). This is the same failure mode quantified earlier as the soft-accuracy versus LLM-judge gap of roughly seven times the corresponding Qwen gap (Table 3). A prediction shown as “(empty)” in Appendix B is the limit case of the same effect: BLIP-2 echoes the prompt without producing new content, so the post-processing that strips the echoed prompt and the trailing **Answer:** marker leaves the empty string.

Taken together, these examples support the quantitative conclusion that closing the remaining accuracy gap on TextVQA needs both better text legibility and stronger question-grounded selection of which visible text to read, while BLIP-2 additionally needs answer-format supervision.

5.6 Summary

The strongest evidence from the experiments is that Qwen TextVQA performance improves through both adaptation and better visual resolution. LoRA fine-tuning provides a clear gain over zero-shot inference, full attention+MLP LoRA outperforms attention-only LoRA, and the high-resolution run produces the best overall validation accuracy. BLIP-2 performs much worse in this setup, which reinforces the model-selection decision to use Qwen as the primary system. The limited effect of OCR-only prompting suggests that the strongest model already understands the instruction to read text, but it still needs either task-specific adaptation or more legible visual inputs to answer reliably.

6 Reproducibility

The complete code for this project, including the data pipeline, model loading, LoRA training, inference, ablation runs, and result generation, is publicly available at <https://github.com/G6KlayWang/TEXTVQA>. The repository contains stage-specific entrypoints, versioned YAML configuration files for each model and run variant, and the scoring code used to produce every metric

reported in this paper, so the numbers and figures here are reproducible end-to-end from a clean clone.

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Appendix

A Training Curve

Figure 1 shows the LoRA training loss and training-time evaluation loss for the main Qwen2.5-VL run, computed against the 512-example holdout from the end of the training split. The curve confirms a stable cosine-scheduled decrease without divergence, and the eval loss tracks the training loss over both epochs, which is consistent with the small generalization gap observed between training-time evaluation and the official 5,000-example validation split.

B Qualitative Examples

Each panel below shows one curated validation example: the source image, the question, the majority human answer, and the prediction from every completed Qwen and BLIP-2 run. Long BLIP-2 outputs that contain repeated phrases are truncated with “...” for readability. A prediction shown as “(empty)” means the cleaned BLIP-2 output is the empty string, which happens when the model echoes the prompt without producing a new answer (see §5.5). The qualitative patterns these examples illustrate are discussed in §5.5.

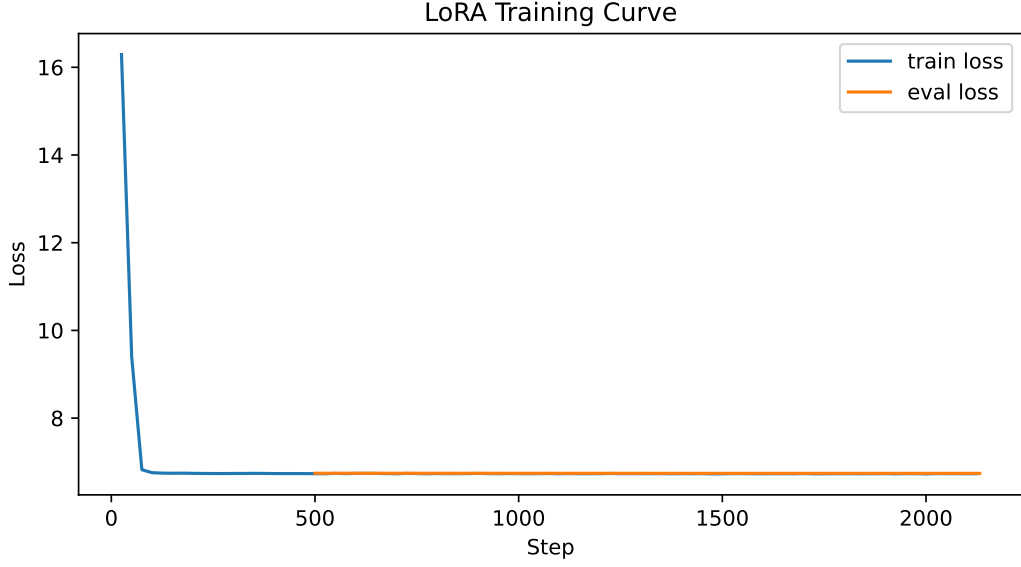


Figure 1: Qwen2.5-VL LoRA training loss and training-time evaluation loss across the two-epoch run. Eval loss is computed on the 512-example holdout taken from the end of the training split.



Q1: What is the brand of the second bottle from the left?
Ground truth: southern comfort

Model	Prediction
Qwen zero-shot	Southern Comfort
Qwen LoRA	southern comfort
Qwen attn-only LoRA	southern comfort
Qwen OCR prompt	Southern Comfort
Qwen OCR hint	Dewar's
Qwen high-res	southern comfort
BLIP-2 zero-shot	whiskey bottle
BLIP-2 LoRA	mccall's mark 70% abv bourbon whiskey
BLIP-2 Q-Former	jack daniels whiskey
BLIP-2 OCR hint	(empty)



Q2: What time does the top clock show?
Ground truth: 10:07

Model	Prediction
Qwen zero-shot	10:25
Qwen LoRA	10:25
Qwen attn-only LoRA	10:35
Qwen OCR prompt	10:25
Qwen OCR hint	10:10
Qwen high-res	10:25
BLIP-2 zero-shot	(empty)
BLIP-2 LoRA	(empty)
BLIP-2 Q-Former	1:00 pm
BLIP-2 OCR hint	(empty)



Q3: What word is written in white at the bottom of the label?

Ground truth: barbieri

Model	Prediction
Qwen zero-shot	Barberi
Qwen LoRA	barberi
Qwen attn-only LoRA	barberi
Qwen OCR prompt	Barberi
Qwen OCR hint	Aperitivo
Qwen high-res	barbieri
BLIP-2 zero-shot	apérool
BLIP-2 LoRA	aperol spumante vermouht di marbella vermouht di marbella. . .
BLIP-2 Q-Former	raspberries
BLIP-2 OCR hint	(empty)



Q4: How much is the pink headband?

Ground truth: \$2.00

Model	Prediction
Qwen zero-shot	100
Qwen LoRA	\$1.99
Qwen attn-only LoRA	unanswerable
Qwen OCR prompt	10
Qwen OCR hint	00228
Qwen high-res	unanswerable
BLIP-2 zero-shot	(empty)
BLIP-2 LoRA	\$1.00 usd.00 usd.00 usd.00. . .
BLIP-2 Q-Former	\$1.00
BLIP-2 OCR hint	(empty)



Q5: What can you not do at anytime?
Ground truth: no standing

Model	Prediction
Qwen zero-shot	STAND
Qwen LoRA	stand
Qwen attn-only	stand
LoRA	
Qwen OCR prompt	STAND
Qwen OCR hint	Stand
Qwen high-res	stand
BLIP-2 zero-shot	(empty)
BLIP-2 LoRA	(empty)
BLIP-2 Q-Former	you can't take a picture of the building
BLIP-2 OCR hint	(empty)



Q6: Which bank is hosting the cup?
Ground truth: charles schwab

Model	Prediction
Qwen zero-shot	Charles Schwab
Qwen LoRA	charles schwab
Qwen attn-only	charles schwab
LoRA	
Qwen OCR prompt	Charles Schwab
Qwen OCR hint	Charles Schwab
Qwen high-res	wells fargo
BLIP-2 zero-shot	(empty)
BLIP-2 LoRA	chase bank of america williams leggings adidas adidas...
BLIP-2 Q-Former	bank of america
BLIP-2 OCR hint	(empty)